

TABLE I
SPI Command Sets (in Hex Format) for Serial Programming
of the Target AVR Microcontroller*

Instruction/operation	SPI instruction format			
	Byte 1	Byte 2	Byte 3	Byte 4
Programming Enable	AC	53	00	00
Chip Erase (program memory/EEPROM)	AC	80	00	00
Poll RDY/BSY	F0	00	00	data byte out
Load instructions				
Load extended address byte	4D	00	Extended adr	00
Load program memory page, high byte	48	adr MSB	adr LSB	high data byte in
Load program memory page, low byte	40	adr MSB	adr LSB	low data byte in
Load EEPROM memory page (page access)	C1	00	adr LSB	data byte in
Read instructions				
Read program memory, high byte	28	adr MSB	adr LSB	high data byte out
Read program memory, low byte	20	adr MSB	adr LSB	low data byte out
Read EEPROM memory	A0	adr MSB	adr LSB	data byte out
Read lock bits	58	00	00	data byte out
Read signature byte	30	00	0000 000aa	data byte out
Read fuse bits	50	00	00	data byte out
Read fuse high bits	58	08	00	data byte out
Read extended fuse bits	50	08	00	data byte out
Read calibration byte	38	00	0b00 000bb	data byte out
Write instructions				
Write program memory page	4C	000a aaaa	aa00 0000	00
Write EEPROM memory	C0	adr MSB	adr LSB	data byte in
Write EEPROM memory page (page access)	C2	adr MSB	adr LSB	00
Write lock bits	AC	E0	00	data byte in
Write fuse bits	AC	A0	00	data byte in
Write fuse high bits	AC	A8	00	data byte in
Write extended fuse bits	AC	A4	00	data byte in

*Commands are sent by the programmer board to the slave device which is being programmed

downloaded for free from IAR website www.iar.com. For details of IAR Embedded Workbench, you may refer to 'A Beginners' Guide to ATMEL AVR Development' article published in January issue of EFY.

Coding for AVR programmer.

When writing serial data to the AVR microcontroller, data is clocked on the rising edge of SCK. When reading data from the AVR microcontroller, data is clocked on the falling edge of SCK. To program and verify the AVR microcontroller in the SPI serial programming mode, the following sequence is recommended (refer Table I for serial

programming command):

1. Power-up: Apply power between V_{CC} and ground (GND) while

RST and SCK are set to '0'. In some systems, the programmer cannot guarantee that SCK is held low during power-up. In this case, RST must be given a positive pulse of at least two CPU clock cycles duration after SCK has been set to '0'.

2. Wait for at least 20 ms and enable SPI serial programming by sending the 'Programming Enable' serial instruction to pin MOSI.

3. The SPI serial programming instructions will not work if the communication is out of synchronisation. When in synchronisation, the second byte (0x53) will echo back when issuing the third byte of the 'Programming Enable' instruction. Whether the echo is correct or not, all four bytes of the instruction must be transmitted. If the 0x53 doesn't echo back, give a positive pulse to reset pin and issue a new 'Programming Enable' command.

4. The Flash is programmed one page at a time. The memory page is loaded one byte at a time by supplying the six least significant bits (LSBs) of the address and data together with the "load program memory page" instruction. To ensure correct loading of the page, the low-byte data must be loaded before high-byte data is applied for a given address. The program memory page is stored by loading the "write program memory page" instruction with the seven most significant bits (MSBs) of the address. If polling is not used, the user must wait for a period of at least t_{WD_FLASH} before issuing the next page. Accessing the SPI serial programming interface before completion of the Flash-write operation can result in incorrect programming.

5. The EEPROM array is programmed one byte at a time by supplying the address and data together with the appropriate write instruction. An EEPROM location is au-

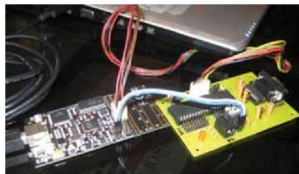


Fig. 5: Burning the hex code into the programmer board microcontroller